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AI-Based E-mail Assistant

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Contents

- Introduction
- Literature survey
- Problem system
- Methodology
- Result and Discussion
- Conclusion and Future Works
- References

Introduction

Email is an important part of daily communication in work, study, and personal life. However, managing many emails is often difficult and time-consuming. The **Smart Email Assistant** uses artificial intelligence to make email handling easier and faster. It works as a Chrome extension with Gmail and helps users write replies, summarize long emails, find important messages, and translate emails into different languages. This system reduces manual effort, saves time, and improves productivity by making email communication smarter and more user-friendly.

Overall, the Smart Email Assistant helps users manage emails more efficiently by reducing workload and saving time. It improves communication quality and allows users to focus on important tasks, making email usage easier, smarter, and more effective in everyday life.

Literature Survey

Many researchers have worked on improving email communication using artificial intelligence. Early work like Intelligent Email (Dredze et al., 2008) used machine learning to predict email replies and detect attachments, which created the base for AI email systems. Later, Thiergart et al. (2021) showed that GPT-3 can be used to write automatic email replies. LaMPost (2022) helped dyslexic users write emails more easily, while Short Mail (2023) focused on summarizing long emails. Randive and Nalkar (2024) studied how AI improves clarity and efficiency in corporate emails. More recent systems such as Miura et al. (2025) and Mail Genius-AI provided complete solutions including email classification, summarization, and formal reply generation, showing continuous improvement in intelligent email assistance.

However, despite these developments, many existing systems still face limitations such as generic replies, limited understanding of context, and privacy concerns. These gaps show the need for a secure, user-friendly, and intelligent email assistant that can provide accurate and reliable support.

Based on this literature survey, it is clear that significant research has been carried out in the field of AI-based email systems. By carefully studying existing approaches and their limitations, this project is designed with a strong understanding of current research, aiming to build a more secure, intelligent, and practical Smart Email Assistant.

Problem System

In today’s professional environment, managing a large number of emails has become a serious problem. Employees receive many emails every day, which creates confusion, stress, and sometimes causes important messages to be missed. This email overload increases mental pressure and reduces work efficiency.

Another major issue is the time required to read and reply to emails. Writing proper and meaningful replies takes a lot of time during the workday. Employees spend a large part of their time reading emails instead of focusing on important tasks.

Existing email systems also lack intelligent support. They do not clearly understand email intent, cannot prioritize important messages, and do not generate smart replies. Because of this, users must manually handle everything. Solving this problem is important because it affects productivity, communication quality, and employee well-being. Therefore, there is a strong need for an automated and intelligent email assistant that can analyze, summarize, and respond to emails efficiently.

Methodology

The proposed Smart Email Assistant uses a microservices-based architecture managed by a Spring Boot backend. This design helps the system work smoothly, securely, and efficiently. Each part of the system has a clear role, which makes the system easy to maintain and scalable.

System Architecture :-

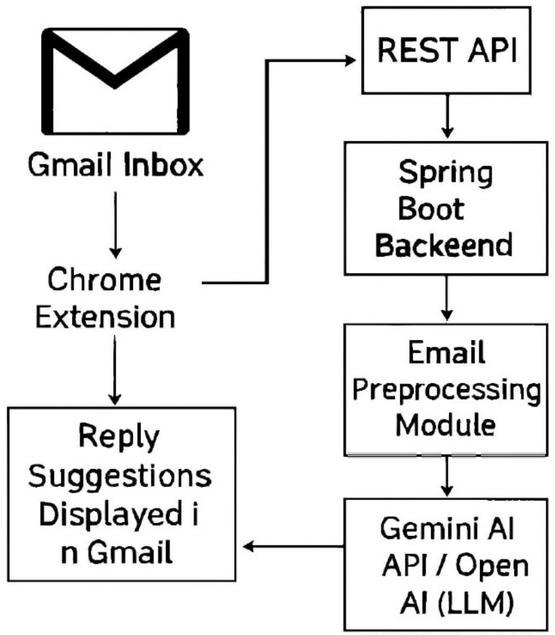
When an email arrives in the Gmail inbox, it is accessed through a Chrome Extension. This extension sends the email data to the Spring Boot backend using REST APIs. The backend manages important tasks such as request handling, data preprocessing, caching, and logging. After this, the backend communicates with the AI inference service (Gemini or OpenAI models) to generate reply suggestions. The generated responses are then sent back and shown directly inside the Gmail interface for the user.

AI Processing Pipeline :-

First, the email text is cleaned and prepared by removing unnecessary words and formatting the content. Next, NLP models like BERT are used to understand the meaning and intent of the email. If the email content is unclear, Gemini AI generates clarifying questions to improve understanding. The system then summarizes the email and adjusts the reply style to match clarity and tone. Finally, GPT and Gemini models generate a natural and human-like reply. Over time, reinforcement learning helps the system improve response quality based on previous interactions.

Methodology: System Architecture Diagram of Smart Email Assistant

This diagram shows the working of the Smart Email Assistant. Emails from the Gmail inbox are accessed through a Chrome extension, which sends the data to a Spring Boot backend using REST APIs. The backend preprocesses the email and sends it to the Gemini/OpenAI model to generate smart reply suggestions. The generated replies are then sent back and displayed directly inside Gmail.



Smart Email Assistant

Result And Discussion

Results :-

- We successfully developed a **Smart Email Assistant** that works directly inside **Gmail** using a Chrome extension.
- The system can **automatically generate smart email replies** based on the email content.
- It can **summarize long email threads** into short and clear points, which saves reading time.
- The assistant helps in **identifying important emails**, so users can focus on priority messages first.
- The system works **smoothly and fast** without affecting Gmail performance.
- Secure login using **OAuth2** keeps user emails and data safe.
- Overall, this project helped in **reducing manual effort** and **improving productivity**.

Discussion :-

- While working on this project, we observed that **AI can make email handling much easier**.
- Writing replies becomes faster because the system suggests meaningful responses.
- Summarization helps users understand long conversations quickly.
- Highlighting important emails reduces confusion and saves time.
- This project shows our effort in combining **AI concepts with real-world problems**.
- The results prove that the system is **useful, practical, and can be improved further** with more features in the future.

Conclusion And Future Work

Conclusion :-

This project shows how artificial intelligence can be used to improve everyday email communication. We developed a Smart Email Assistant that works as a Chrome extension inside Gmail using React, a Java backend, and the Gemini API. The assistant helps users by suggesting smart replies, summarizing long email threads, and highlighting important emails. From testing and analysis, I observed that the system reduces manual work, saves time, and improves clarity while writing and reading emails.Secure login using OAuth2 ensures that user data remains safe. Overall, this project successfully meets its goal of making email management smarter, faster, and more user-friendly. It also reflects my effort in combining AI concepts with real-world problems using modern web technologies.

Future Work :-

*Although the system works well, there are many ways it can be improved in the future :-

- Sentiment analysis can be added to adjust the tone of emails (formal, polite, or friendly).
- Voice-to-text feature can help users write emails faster by speaking instead of typing.
- Calendar integration can be included to support smart meeting scheduling from emails.
- The assistant can be extended to mobile platforms for use on smartphones.
- Better email categorization (work, personal, urgent) can further improve organization.
- Personalization features can be added to learn user writing style over time.
- These future improvements will make the Smart Email Assistant more powerful, intelligent, and useful in real-life communication.

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THANK YOU